

Instructions for Subjective Well-Being Exercise

I. INTRODUCTION

In this exercise, you will explore the question "Does money buy happiness?" More precisely, you will investigate whether, in a sample of low- to middle-income countries, there is any association between a country's income per capita and the average level of overall life satisfaction reported by residents of the country.

I.A. The data

The data for this exercise are taken from two sources, the World Bank's *World Development Indicators* (WDI),¹ and a study titled *Summer 2002 44 Nation Survey* conducted by the Pew Research Center.²

The WDI data are provided to you in a file named *wdi_input.csv*. That file contains a spreadsheet with 195 observations, each representing a country, and two variables: *country*, which gives the name of the country, and *gdppc*, which gives the country's 2002 income per capita in purchasing power parity dollars.

The data from the Pew study are in a file named *pew_input.csv*. The spreadsheet in that file includes 33,746 observations, each representing one respondent to the 44 nation survey. The variable *country* identifies the respondent's country of residence. The variable *swb* records the respondent's subjective assessment of where they stand on a "ladder of life" with 11 steps, from 0, representing "the worst possible life for you", to 10, representing the "best possible life for you." Questions like this one, that ask respondents to report on some scale their overall quality of life or level of satisfaction, are known as measures of "subjective well-being" (SWB).

I.B Brief overview of the exercise

Your main task is to construct a scatterplot of the average subjective well-being of respondents in each country in the sample against the countries' incomes per capita,

¹ <https://databank.worldbank.org/source/world-development-indicators>

² <https://www.pewresearch.org/global/2002/10/31/summer-2002-survey-data/>

and then to comment on what that figure suggests about the association between income and well-being.

Before you can construct that scatterplot, you need to merge the Pew data, which has information on subjective well-being, with the WDI data, which has information on incomes per capita. But the datasets you start with (*wdi_input.csv* and *pew_input.csv*) need to be modified before they can be merged.

The original Pew data, which has one observation for each individual survey respondent, must be collapsed to a dataset that has one observation per country, and a new variable equal to the mean of the values of *swb* reported by all respondents in each country. This new variable will be called *meanswb*. After you have made these changes to the Pew data, you will save it in a Stata-formatted data file called *pew_intermediate.dta*.

The original WDI data includes observations for countries at all income levels. Because this exercise focuses on low- and middle-income countries, you will remove the observations for all countries with incomes per capita greater than 15,000 purchasing power parity dollars. You will save this modified data in a new data file called *wdi_intermediate.dta*.

You will then merge the two intermediate data files, *pew_intermediate.dta* and *wdi_intermediate.dta*, and save the combined data in a file called *analysis.dta*. You will use the data in *analysis.dta* to produce some descriptive statistics and figures for the variables *meanswb* and *gdppc*, and to generate the main result of this exercise, the scatterplot of *meanswb* against *gdppc*.

Finally, you will write a report in which you present and comment on the summary statistics and scatterplot described above.

I.C. Reproduction documentation

You will also assemble a package of materials referred to as **reproduction documentation**. These materials will include a set of folders and subfolders containing all the data files, scripts, and output that were used in or generated by the computations you carried out for this exercise. This documentation will contain everything necessary

to make it possible for your instructor (or any third party) to reproduce all the steps of data processing and analysis you carried out to generate the results presented in your report—simply by running your scripts.

The keys to constructing effective reproduction documentation lie in a set of concepts known as the ***reproducibility trifecta***:

- You must keep all your data, scripts, and output files in a hierarchy of folders with a fixed structure.
- You must be attentive to which folder is designated as Stata's working directory whenever one of your scripts is executed.
- In your scripts, when you need to specify the location of a particular folder or file, you must do so using a relative directory path, from whichever folder is designated as the working directory to the target folder or file.

These principles are embodied in the detailed instructions for constructing reproduction documentation throughout this exercise.

II. DETAILED INSTRUCTIONS

II.A. Build a directory structure.

A directory structure is simply a hierarchy, or tree, of folders. You will build the directory structure you need for this project in three steps:

- Create a project folder.
- Build a hierarchy of folders and subfolders inside the project folder.
- Put copies of the two data files you were given for this project in a particular folder in your hierarchy.

II.A.1. Create a project folder.

→ Make a new folder on your computer, and give it the name **SWBExerciseI**.

SWBExerciseI will serve as your ***project folder***. It is where you will store all the files you work with for this exercise.

Keep this folder in a convenient place on your computer.

II.A.2. Build a directory structure inside the project folder.

→ In **SWBExercise/**, build a hierarchy of folders, as illustrated below:

```
SWBExercise/
├── Data/
│   ├── InputData/
│   ├── IntermediateData/
│   └── AnalysisData/
├── Scripts/
│   ├── ProcessingScripts/
│   └── AnalysisScripts/
└── Output/
    ├── Figures/
    └── Tables/
```

Throughout your work on this exercise, each time you obtain or generate a new file, you will store it in the appropriate folder in this directory structure.

When you have completed the exercise, this directory structure and the files you have stored in it will constitute your reproduction documentation.

II.A.3. Put the input data files in the **InputData/** folder.

Data files you originally obtain for a project and have not yet modified in any way are called **input data files**. For this exercise, there are two input data files: *pew_input.csv* and *wdi_input.csv*.

→ Put copies of the two input data files in the **InputData/** folder.

II.B. Install a user-written Stata package.

At one point in this exercise, you will need a user-written Stata package called "asdoc".

To install asdoc, launch the copy of Stata you will be using for the exercise, and in the command line type:

```
ssc install asdoc
```

Exit Stata after you have installed the asdoc package.

II.C. Write three processing scripts.

The input data files you are given for this exercise need to be modified in a number of ways before you can perform the analysis you will be doing for this exercise. The steps required to prepare the input data for analysis are called **processing**.

You will write three scripts with commands that execute all the steps of processing necessary for this exercise. Since you will be using Stata, these scripts are called do-files, and their filename extension is *.do*.

- *pew_processing.do*: Prepares the Pew data for merging with the WDI data.
- *wdi_processing.do*: Prepares the WDI data for merging with the Pew data.
- *merging.do*: Merges the Pew and WDI data, and saves the data in a new file that you will use when you conduct the analysis.

II.C.1. Write the *pew_processing.do* script.

In *pew_processing.do*, you will write Stata commands that prepare the Pew input data for merging with the WDI data.

You will write this script in four stages:

- Launching Stata, and setting the working directory.
- Setting up a new do-file.
- Writing comments and commands in the do-file.
- Executing and saving the do-file.

Launching Stata and setting the working directory.

→ Launch Stata.

→ Set the working directory to the project folder (**SWBExerciseI**).

When you set the working directory, do so interactively, not by writing a command in the script that sets the working directory.

Setting up the *pew_processing.do* script.

→ Open a new do-file, and save it in the **ProcessingScripts/** folder, giving it the name *pew_processing.do*.

→ At the top of this do-file, write a comment indicating that, before executing this script, the user should ensure that the project folder (**SWBExercise/**) is designated as Stata's working directory.

Here is an example of a comment that would serve this purpose:

```
/* Whenever this do-file is being edited or executed, the
   project folder (SWBExercise/) should be designated as
   Stata's working directory. */
```

Writing comments and commands in *pew-processing.do*.

→ Immediately after the comment about the working directory, write a series of commands that accomplish the tasks listed below.

Important: Before every command, insert a comment explaining the purpose of the command.

- Clear Stata's memory.

Stata syntax:

```
clear
```

- Read the Pew input data from *pew_input.csv*. Use a relative directory path to indicate that *pew_input.csv* is stored in the **InputData/** folder.

Stata syntax:

```
import delimited using Data/InputData/pew_input.csv
```

- Collapse the individual-level Pew input data into country-level data, in which the values of *swb* are equal to the means of the individual values for each country.

Stata syntax:

```
collapse swb, by(country)
```

- Because the variable *swb* in the collapsed data gives country means of the individual responses in the original data, change the name of *swb* to *meanswb*.

Stata syntax:

```
rename swb meanswb
```

- Assign the label "Mean SWB" to the *meanswb* variable.

Stata syntax:

```
label variable meanswb "Mean SWB"
```

- Save the modified data in a new file called *pew_intermediate.dta*. Use a relative directory path to indicate that this file should be saved in the **IntermediateData/** folder.

Stata syntax:

```
save Data/IntermediateData/pew_intermediate.dta, replace
```

pew_intermediate.dta contains data you have modified, but needs further processing before it can be used in your analysis. It is therefore called an **intermediate data file**.

Executing and saving *pew-processing.do*.

→ After you have finished writing the commands described above, execute *pew_processing.do* from start to finish, and be sure that it runs without causing an error.

→ Check that when you execute *pew_processing.do* it generates the *pew_intermediate.dta* data file and saves it in the **IntermediateData/** folder.

→ Save the complete *pew_processing.do* script in the **ProcessingScripts/** folder (which is where you saved it when you first created it).

II.C.2. Write the *wdi_processing.do* script.

In *wdi_processing.do*, you will write Stata commands that prepare the WDI input data for merging with the Pew data.

You will write this script in four stages:

- Launching Stata, and setting the working directory.
- Setting up a new do-file.
- Writing comments and commands in the do-file.
- Executing and saving the do-file.

Launching Stata and setting the working directory.

The instructions for launching Stata and setting the working directory when you are preparing to write the *wdi_processing.do* script are exactly the same as they were when you were preparing to write the *pew_processing.do* script. See section II.C.1 for details.

The key thing is to be sure that after launching Stata you set the project folder (**SWBExerciseI**) as the working directory.

Setting up the *wdi_processing.do* script.

→ Open a new do-file, and save it in the **ProcessingScripts/** folder, giving it the name *wdi_processing.do*.

→ At the top of this do-file, write a comment indicating that, before executing this script, the user should ensure that the project folder (**SWBExerciseI**) is designated as Stata's working directory.

A comment identical to the one you wrote at the beginning of *pew_processing.do* would also be appropriate at the beginning of *wdi_processing.do*. See section II.C.1 for details.

Writing comments and commands in *wdi_processing.do*.

→ Immediately after the comment about the working directory, write a series of commands that accomplish the tasks listed below.

Important: Before every command, insert a comment explaining the purpose of the command.

- Clear Stata's memory.

Stata syntax:
`clear`

- Read the WDI input data from *wdi_input.csv*. Use a relative directory path to indicate that *wdi_input.csv* is stored in the **InputData/** folder.

Stata syntax:
`import delimited using Data/InputData/wdi_input.csv`

- Assign the label "GDP pc (\$PPP)" to the variable *gdppc*.

Stata syntax:
`label variable gdppc "GDP pc ($PPP)"`

- Because this exercise is an exploration of the association between income per capita and subjective well-being in low- and middle-income countries, we drop the observations for upper-income countries, which we define as countries with 2002 GDP per capita of more than \$PPP 15,000.

Stata syntax:
`drop if gdppc>15000`

- Save the modified data in a new file called *wdi_intermediate.dta*. Use a relative directory path to indicate that this file should be saved in the **IntermediateData/** folder.

Stata syntax:

```
save Data/IntermediateData/wdi_intermediate.dta, replace
```

wdi_intermediate.dta contains data you have modified, but needs further processing before it can be used in your analysis. It is therefore called an **intermediate data file**.

Executing and saving *wdi-processing.do*.

→ After you have finished writing the commands described above, execute *wdi_processing.do* from start to finish, and be sure that it runs without causing an error.

→ Check that when you execute *wdi_processing.do* it generates the *wdi_intermediate.dta* data file and saves it in the **IntermediateData/** folder.

→ Save the complete *wdi_processing.do* script in the **ProcessingScripts/** folder (which is where you saved it when you first created it).

II.C.3. Write the *merging.do* script.

In *merging.do*, you will write Stata commands that merge the intermediate Pew data with the intermediate WDI data.

You will write this script in four stages:

- Launching Stata, and setting the working directory.
- Setting up a new do-file.
- Writing comments and commands in the do-file.
- Executing and saving the do-file.

Launching Stata and setting the working directory.

The instructions for launching Stata and setting the working directory when you are preparing to write the *merging.do* script are exactly the same as they were when you were preparing to write the *pew_processing.do* script. See section II.C.1 for details.

The key thing is to be sure that after launching Stata you set the project folder (**SWBExerciseI**) as the working directory.

Setting up the *merging.do* script.

→ Open a new do-file, and save it in the **ProcessingScripts/** folder, giving it the name *merging.do*.

→ At the top of this do-file, write a comment indicating that, before executing this script, the user should ensure that the project folder (**SWBExerciseI**) is designated as Stata's working directory.

A comment identical to the one you wrote at the beginning of *pew_processing.do* would also be appropriate at the beginning of *merging.do*. See section II.C.1 for details.

Writing comments and commands in *merging.do*.

→ Immediately after the comment about the working directory, write a series of commands that accomplish the tasks listed below.

Important: Before every command, insert a comment explaining the purpose of the command.

- Clear Stata's memory.

Stata syntax:
`clear`

- Read the intermediate Pew data from *pew_intermediate.dta*. Use a relative directory path to indicate that *pew_intermediate.dta* is stored in the **IntermediateData/** folder.

Stata syntax:
`use Data/IntermediateData/pew_intermediate.dta`

- Merge the intermediate Pew data (which was just read into memory) with the intermediate WDI data in *wdi_intermediate.dta* (which is stored in the **IntermediateData/** folder), matching on the variable *country*. Use a relative directory path to indicate that *wdi_intermediate.dta* is stored in the **IntermediateData/** folder.

Stata syntax:
`merge 1:1 country ///
using Data/IntermediateData/wdi_intermediate.dta`

- Keep only the observations for countries that appear in both the Pew and WDI data (i.e., countries that matched when the merge was performed).

Stata syntax:
`keep if _merge==3`

- Drop the *_merge* variable that was created when the merge command was executed.

- Save the modified data in a new file called *analysis.dta*. Use a relative directory path to indicate that this file should be saved in the **AnalysisData/** folder.

Stata syntax:

```
save Data/AnalysisData/analysis.dta, replace
```

analysis.dta contains data on which all the necessary processing has been carried out and is ready to be used in your analysis. It is therefore called an **analysis data file**.

Executing and saving *merge.do*.

→ After you have finished writing the commands described above, execute *merge.do* from start to finish, and be sure that it runs without causing an error.

→ Check that when you executed *merge.do* it generated the *analysis.dta* data file and saved it in the **AnalysisData/** folder.

→ Save the complete *merge.do* script in the **ProcessingScripts/** folder (which is where you saved it when you first created it).

II.D. Write an analysis script.

The **analysis** for this project consists of:

- Generating a table of descriptive statistics for the variables *meanswb* and *gdppc*.
- Generating a histograms of *meanswb* and *gdppc*.
- Generating a scatterplot of *meanswb* against *gdppc*.

You will write a script, called *analysis.do*, containing Stata commands that open the analysis data file (*analysis.dta*), generate the table, histograms, and scatterplot, and save them in the appropriate folders in your directory structure.

You will write this script in four stages:

- Launching Stata, and setting the working directory.
- Setting up a new do-file.
- Writing comments and commands in the do-file.
- Executing and saving the do-file.

Launching Stata and setting the working directory.

The instructions for launching Stata and setting the working directory when you are preparing to write the *analysis.do* script are exactly the same as they were when you were preparing to write the *pew_processing.do* script. See section II.C.1 for details.

The key thing is to be sure that after launching Stata you set the project folder (**SWBExercise/**) as the working directory.

Setting up the *analysis.do* script.

→ Open a new do-file, and save it in the **AnalysisScripts/** folder, giving it the name *analysis.do*.

→ At the top of this do-file, write a comment indicating that, before executing this script, the user should ensure that project folder (**SWBExercise/**) is designated as Stata's working directory.

A comment identical to the one you wrote at the beginning of *pew_processing.do* would also be appropriate at the beginning of *analysis.do*. See section II.C.1 for details.

Writing comments and commands in *analysis.do*.

→ Immediately after the comment about the working directory, write a series of commands that accomplish the tasks listed below.

Important: Before every command, insert a comment explaining the purpose of the command.

- Clear Stata's memory.

Stata syntax:
`clear`

- Read the analysis data from *analysis.dta*. Use a relative directory path to indicate that *analysis.dta* is stored in the **AnalysisData/** folder.

Stata syntax:
`use Data/AnalysisData/analysis.dta`

- Generate a table showing summary statistics for the two variables *meanswb* and *gdppc*, and save the table in the **Tables/** folder, in a rich text format file named *Table1.rtf*.

The summary statistics for each variable should include:

- The sample mean and standard deviation.
- The 25th, 50th, and 75th percentiles.

To save the table to a rich text file, use the user-written *asdoc* prefix at the beginning of the command. (For that to be possible, you must have installed the *asdoc* package, as you were instructed to do in section II.B.)

Use a relative directory path to indicate that the table should be saved in the **Tables/** folder.

Stata syntax:

```
asdoc tabstat meanswb gdppc, ///
      statistics(mean sd p25 p50 p75) ///
      save(Output/Tables/Table1.rtf) ///
      replace
```

- Generate a histogram of *meanswb*, and save the histogram in the **Figures/** folder, in a graphics file named *Figure1.gph*.

Use a relative directory path to indicate that *Figure1.gph* should be saved in the **Figures/** folder.

Stata syntax:

```
histogram meanswb, percent ///
      saving(Output/Figures/Figure1.gph, replace)
```

- Generate a histogram of *gdppc*, and save the histogram in the **Figures/** folder, in a graphics file named *Figure2.gph*.

Use a relative directory path to indicate that *Figure2.gph* should be saved in the **Figures/** folder.

Stata syntax:

```
histogram gdppc, percent ///
      saving(Output/Figures/Figure2.gph, replace)
```

- Generate a scatterplot of *gdppc* (on the horizontal axis) and *meanswb* (on the vertical axis), showing the line of best fit. Save the scatterplot in the **Figures/** folder, in a graphics file named *Figure3.gph*.

Use a relative directory path to indicate that *Figure3.gph* should be saved in the **Figures/** folder.

Stata syntax:

```
scatter meanswb gdppc, ///
      saving(Output/Figures/Figure3.gph, replace)
```

Executing and saving *analysis.do*.

→ After you have finished writing the commands described above, execute *analysis.do* from start to finish, and be sure that it runs without causing an error.

→ Check that when you executed *analysis.do* it generated the table (*Table1.rtf*) and the three figures (*Figure1.gph*, *Figure2.gph*, and *Figure3.gph*) and saved them in the appropriate folders (**Tables/** or **Figures/**).

→ Save the complete *analysis.do* script in the **AnalysisScripts/** folder (which is where you saved it when you first created it).

II.E Write an all-in-one script.

You will give your *all-in-one script* the name *aio.do*. It will contain commands that tell Stata to execute the four scripts you have written—the three processing scripts and the analysis script. So it will be possible to reproduce all your work on this exercise simply by running *aio.do*.

You will write the *aio.do* script in four stages:

- Launching Stata, and setting the working directory.
- Setting up a new do-file.
- Writing comments and commands in the do-file.
- Executing and saving the do-file.

Launching Stata and setting the working directory.

The instructions for launching Stata and setting the working directory when you are preparing to write the *aio.do* script are exactly the same as they were when you were preparing to write the *pew_processing.do* script. See section II.C.1 for details.

The key thing is to be sure that after launching Stata you set the project folder (**SWBExercise/**) as the working directory.

Setting up the *aio.do* script.

→ Open a new do-file, and save it in the top level of your **Scripts/** folder, giving it the name *aio.do*.

→ At the top of this do-file, write a comment indicating that, before executing this script, the user should ensure that project folder (**SWBExercise/**) is designated as Stata's working directory.

A comment identical to the one you wrote at the beginning of *pew_processing.do* would also be appropriate at the beginning of *aio.do*. See section II.C.1 for details.

Writing comments and commands in *aio.do*.

→ Immediately after the comment about the working directory, write a series of commands that accomplish the tasks listed below.

Important: Before every command, insert a comment explaining the purpose of the command.

- Execute *pew_processing.do*.

Use a relative directory path to indicate that *pew_processing.do* is stored in the **ProcessingScripts/** folder.

Stata syntax:

```
do Scripts/ProcessingScripts/pew_processing.do
```

- Execute *wdi_processing.do*.

Use a relative directory path to indicate that *wdi_processing.do* is stored in the **ProcessingScripts/** folder.

Stata syntax:

```
do Scripts/ProcessingScripts/wdi_processing.do
```

- Execute *merge.do*.

Use a relative directory path to indicate that *merge.do* is stored in the **ProcessingScripts/** folder.

Stata syntax:

```
do Scripts/ProcessingScripts/merge.do
```

- Execute *analysis.do*.

Use a relative directory path to indicate that *analysis.do* is stored in the **AnalysisScripts/** folder.

Stata syntax:

```
do Scripts/AnalysisScripts/analysis.do
```

Executing and saving *aio.do*.

→ After you have finished writing the commands described above, execute *aio.do* from start to finish, and be sure that it runs without causing an error.

→ Check that when you executed *aio.do*:

- It generated the *pew_intermediate.dta* data file, and saved it in the **IntermediateData/** folder.
- It generated the *wdi_intermediate.dta* data file, and saved it in the **IntermediateData/** folder.
- It generated the *analysis.dta* data file, and saved it in the **AnalysisData/** folder.
- It generated the *Table1.rtf* rich text file, and saved it in the **Tables/** folder.
- It generated the *Figure1.gph*, *Figure2.gph*, and *Figure3.gph* graphics files, and saved them in the **Figures/** folder.

→ Save the complete *aio.do* script in the top level of the **Scripts/** folder (which is where you saved it when you first created it).

II.F. Write a read me file.

The read me file is a message from you to anyone interested in exploring your documentation and/or reproducing your work for the exercise by running your scripts. Its purpose is to give the user a guide to the contents of your reproduction documentation, and explain how the documentation can be used to reproduce your results.

For this exercise, the read me file should consist of three sections:

- A map of your reproduction documentation.
- Information on the software and hardware you used for the exercise.
- Step by step instructions for reproducing the computations you executed to generate your results.

→ Create a new document with the title "Read Me File for Subjective Well-Being Exercise", with your name and the date at the top. You may create this document with any word processing or typesetting software you like; whatever format you create it in, you will save the final version in *.pdf* format.

→ Write the three sections of the read me file:

Section 1: Contents of the reproduction documentation

Provide a map or a tree that illustrates the directory structure of your reproduction documentation. The illustration of the final directory structure for this exercise (see the beginning of section IV below) would serve this purpose.

Section 2: Software and operating system

State that all computations for this exercise were performed with Stata, and indicate the number of the version and the "flavor" (e.g., BE, SE, or MP) you used.

Indicate that one of your do-files (*analysis.do*) uses a command from the user-written package "asdoc", and that there is a command at the beginning of the do-file that installs this package.

Also state the operating system you used (e.g., Windows, MacOS, or Linux), and the version number or name.

Section 3: Instructions for using your documentation to reproduce your exercise

Explain in plain English the steps the user must take to reproduce your work on the exercise.

Essentially, these instructions boil down to "Put a copy of the reproduction documentation on your computer, and then run the scripts, in this order..."

But the instructions you write in your read me file should be more detailed and explicit. For example, if the reproduction documentation comes in a zipped folder, it would be worth mentioning that it needs to be unzipped. It would probably also be helpful to add that instead of running all the scripts in a certain order, it is possible just to run the all-in-one script, which then calls all the others.

Be sure that the instructions you write are clear and concrete enough that it is realistic to expect that a user who follows them carefully step-by-step will actually be able to succeed in running all your scripts and reproducing your results.

Important: In these instructions, any time you indicate that someone should execute a script, you should say which folder should be designated as the working directory when they do so. (And given the instructions for this exercise, that should be easy—whenever any script is executed the working directory should be set to the project folder, **SWBExercise/**.)

→ When you have finished writing the read me, save it in *.pdf* format, with the name *ReadMe.pdf*, and put it in the top level of your project folder (**SWBExercise/**).

III. THE REPORT

In the report, you will present the table and the three figures that were produced by your analysis, and comment on what you see in them.

→ Create a new document with the title "Report for Subjective Well-Being Exercise", with your name and the date at the top. You may create this document with any word

processing or typesetting software you like; whatever format you create it in, you will save the final version in *.pdf* format.

→ At the beginning of the document, paste in a copy of the table of descriptive statistics for the variables *meanswb* and *gdppc*. (This table should be saved in a file called *Table1.rtf* that was created and stored in your **Tables/** folder when you executed your *analysis.do* script.)

Label the table with an informative title. (Here is an example of an informative title for the table: "Table of Descriptive Statistics for Mean SWB and Income per capita".)

→ Next, paste in copies of the histograms you created for the variables *meanswb* and *gdppc*. (These histograms should be saved in files called *Figure1.gph* and *Figure2.gph* that were created and stored in your **Figures/** folder when you executed your *analysis.do* script.)

Label each figure with an informative title.

→ Paste in a copy of the scatterplot of *gdppc* against *meanswb*. (This scatterplot should be saved in a file called *Figure3.gph* that was created and stored in your **Figures/** folder when you executed your *analysis.do* script.)

Label the scatterplot with an informative title.

→ Write a brief paragraph on what the scatterplot suggests about the association between income per capita and average subjective well-being in low-income countries.

→ When you have finished writing the report, save it in *.pdf* format, with the name *TheReport.pdf*, and put it in the top level of your project folder (**SWBExerciseI**).

IV. TURNING IN YOUR WORK

When you have finished the exercise, your project folder and directory structure should match the illustration on the following page:

SWBExercise/

- |—*TheReport.pdf*
- |—*ReadMe.pdf*
- |—**Data/**
 - | |—**InputData/**
 - | |—*pew_input.csv*
 - | |—*wdi_input.csv*
 - | |—**IntermediateData/**
 - | |—*pew_intermediate.dta*
 - | |—*wdi_intermediate.dta*
 - | |—**AnalysisData/**
 - | |—*analysis.dta*
- |—**Scripts/**
 - | |—*aio.do*
 - | |—**ProcessingScripts/**
 - | |—*pew_processing.do*
 - | |—*wdi_processing.do*
 - | |—*merge.do*
 - | |—**AnalysisScripts/**
 - | |—*analysis.do*
- |—**Output/**
 - | |—**Figures/**
 - | |—*Figure1.gph*
 - | |—*Figure2.gph*
 - | |—*Figure3.gph*
 - | |—**Tables/**
 - | |—*Table1.rtf*

→ Zip your project folder to create a zipped folder named **SWBExercise.zip/**.

→ Upload **SWBExercise.zip/** to the course management system or file sharing platform being used for this course.